

Euclidean Space

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1 Euclidean Spaces

Let E be finite-dimensional Euclidean space endowed with a positive definite symmetric bilinear operator $\langle \cdot, \cdot \rangle$. Let $\mathcal{A} : E \rightarrow E$ be a linear operator, so $\varphi_{\mathcal{A}} : E \times E \rightarrow \mathbb{R}$ given by $\varphi_{\mathcal{A}}(x, y) = \langle x, \mathcal{A}y \rangle$ is a bilinear operator.

Fix an orthonormal basis $\{e_i\}$ of E . The matrix of $\varphi_{\mathcal{A}}$ is the matrix of $\mathcal{A}$ in $\{e_i\}$.

Proof: $\langle e_i, \mathcal{A}e_j \rangle = a_{ij}$ for all $i, j \in [n]$, so the result follows.

Corr: $\varphi_{\mathcal{A}}$ is nondegenerate iff $\mathcal{A}$ is nondegenerate.

Suppose for all $x \in E$, we have $\langle \mathcal{A}x, \mathcal{A}x \rangle = \langle x, x \rangle$. Then $\langle \mathcal{A}x, \mathcal{A}y \rangle = \langle x, y \rangle$ for all $x, y \in E$.

Proof: Observe

$$\begin{aligned} \langle \mathcal{A}(x+y), \mathcal{A}(x+y) \rangle &= \langle \mathcal{A}x + \mathcal{A}y, \mathcal{A}x + \mathcal{A}y \rangle \\ &= \langle \mathcal{A}x, \mathcal{A}x \rangle + 2\langle \mathcal{A}x, \mathcal{A}y \rangle + \langle \mathcal{A}y, \mathcal{A}y \rangle = \langle x, x \rangle + 2\langle \mathcal{A}x, \mathcal{A}y \rangle + \langle y, y \rangle. \end{aligned}$$

Also,

$$\langle \mathcal{A}(x+y), \mathcal{A}(x+y) \rangle = \langle x+y, x+y \rangle = \langle x, x \rangle + 2\langle x, y \rangle + \langle y, y \rangle,$$

so $\langle \mathcal{A}x, \mathcal{A}y \rangle = \langle x, y \rangle$, as desired. Suppose $\varphi_{\mathcal{A}}$ equals $\langle \cdot, \cdot \rangle$. Then $\mathcal{A} = \mathcal{E}$.

Proof: Fix an orthonormal basis $\{e_i\}$ of E . The matrix of $\mathcal{A}$ in $\{e_i\}$ equals the matrix of $\varphi_{\mathcal{A}}$, which equals the matrix of $\langle \cdot, \cdot \rangle$ in $\{e_i\}$, which is I_n . It follows that $\mathcal{A} = \mathcal{E}$.

Claim: Let $\mathcal{A} : E \rightarrow \mathcal{E}$ be a distance preserving linear operator, ie for all $x \in E$, we have $\langle \mathcal{A}x, \mathcal{A}x \rangle = \langle x, x \rangle$. Then $\mathcal{A}$ is orthogonal, ie $\mathcal{A}^* \mathcal{A} = \mathcal{A} \mathcal{A}^* = \mathcal{E}$.

Proof: From before, for all $x, y \in E$, we have $\langle x, \mathcal{A}^* \mathcal{A}y \rangle = \langle \mathcal{A}x, \mathcal{A}y \rangle = \langle x, y \rangle$, so $\mathcal{A}^* \mathcal{A} = \mathcal{E}$. Since E is finite dimensional, $\mathcal{A} \mathcal{A}^* = \mathcal{E}$ as well.

2 Automorphisms of the Quaternions

Suppose $\rho : \mathbb{H} \rightarrow \mathbb{H}$ is bijective and for all $X, Y \in \mathbb{H}, \lambda \in \mathbb{R}$ that

$$\begin{aligned}\rho(X + Y) &= \rho(X) + \rho(Y) \\ \rho(\lambda X) &= \lambda \rho(X) \\ \rho(XY) &= \rho(X)\rho(Y).\end{aligned}$$

Then $\rho(\overline{X}) = \overline{\rho(X)}$ and $N(\rho(X)) = N(X)$.

Proof: Consider in general $X \in \mathbb{H}$ such that $X^2 = -1$. Then $X(-X) = 1$ and $N(X)^2 = N(X^2) = N(-1) = 1$, so $N(X) = 1$ and $X^{-1} = \overline{X}$. It follows $\overline{X} = -X$, or $X + \overline{X} = 0$.

Conversely, suppose $X + \overline{X} = 0$ and $N(X) = 1$. Then $X^2 + \overline{X}X = (X + \overline{X})X = 0$, so $X^2 = -1$.

Observe that $\rho(0) = 0$ and $\rho(1)^2 = \rho(1) \rightarrow \rho(1) = 0, 1$. Since ρ is bijective, $\rho(1) = 1$. It follows by the second condition that $\rho(\lambda) = \lambda$ for all $\lambda \in \mathbb{R}$. Hence, $\rho(i)^2 = \rho(i^2) = \rho(-1) = -1$, so $\rho(i) + \overline{\rho(i)} = 0$. Similar results hold for j, k .

Now for $X = a + bi + cj + dk$,

$$\begin{aligned}\rho(\overline{X}) &= \rho(a - bi - cj - dk) = a - b\rho(i) - c\rho(j) - d\rho(k) \\ &= \overline{\rho(a)} + b\overline{\rho(i)} + c\overline{\rho(j)} + d\overline{\rho(k)} = \overline{\rho(a) + b\rho(i) + c\rho(j) + d\rho(k)} \\ &= \overline{\rho(a + bi + cj + dk)} = \overline{\rho(X)}.\end{aligned}$$

It follows that

$$N(\rho(X)) = \rho(X)\overline{\rho(X)} = \rho(X)\rho(\overline{X}) = \rho(X\overline{X}) = \rho(N(X)) = N(X),$$

as desired.

3 Isometries

Let E be finite-dimensional Euclidean space. Suppose $h, k : E \rightarrow E$ are isometries. Then $kh : E \rightarrow E$ is an isometry.

Proof: For all $x, y \in E$, $\|kh(x) - kh(y)\| = \|h(x) - h(y)\| = \|x - y\|$.